Turbine Cooling

AERSP 410 – Aerospace Propulsion

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Lectures #40–41
December 6–9, 2024

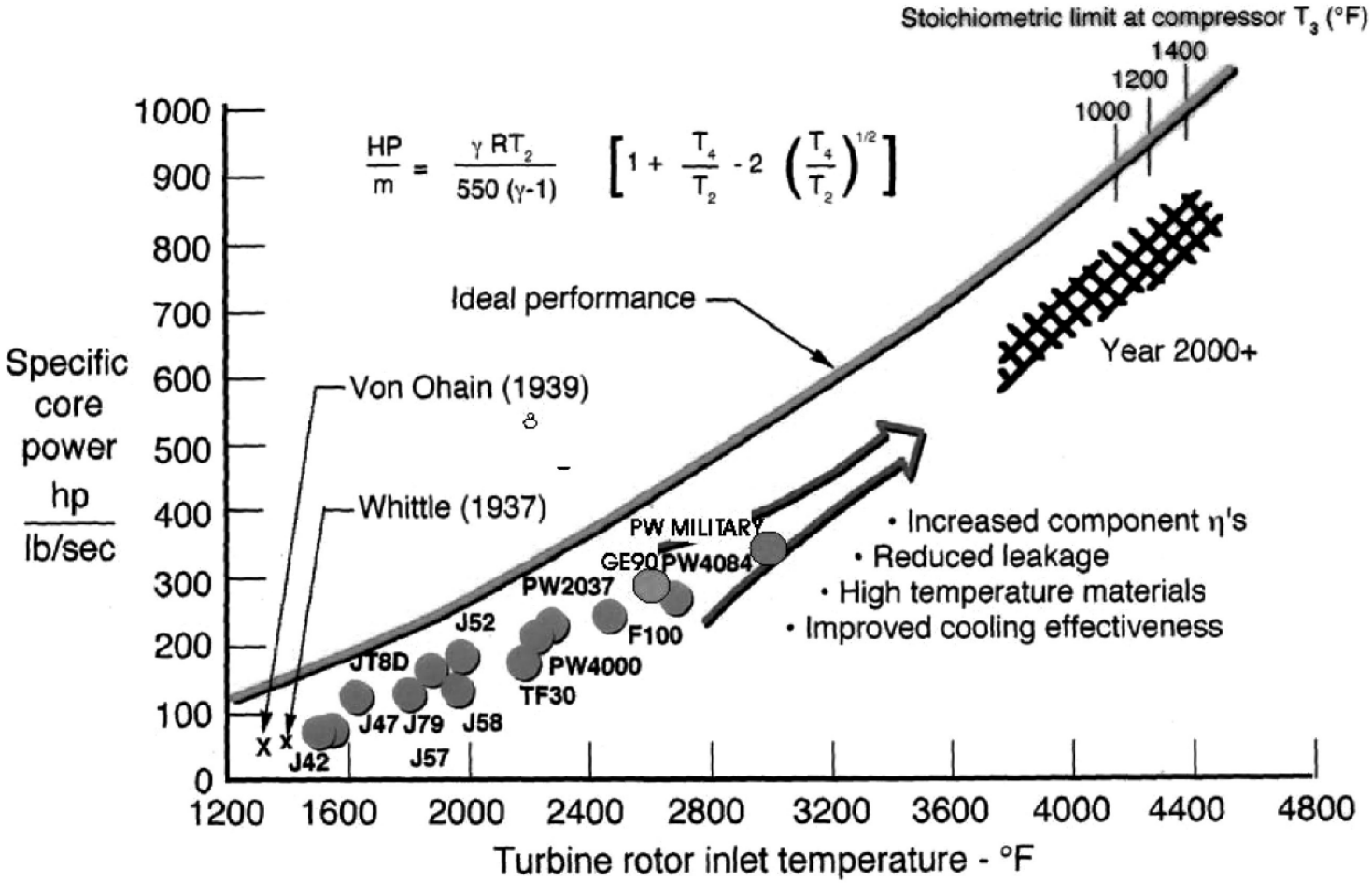
Reminders

- Suggested **reading**
  - Hill & Peterson Section 8.5
- **Homework #8** due before class on Friday
- What was the **muddiest points** of the lecture?
  - Post to discussion thread on Canvas

Main Messages

- Higher turbine inlet temperatures $\longrightarrow$ higher specific thrust, lower weight
- Allowable blade and disc stresses decrease as their temperatures increase
- **Turbine cooling** can keep material temperatures well below hot gas temperatures via continuous heat transfer
  - Convective heat transfer from blade to a cooling flow
  - Thermal barrier coatings with low thermal conductivity

Lighter Engines Enabled by Higher Cycle Temperatures

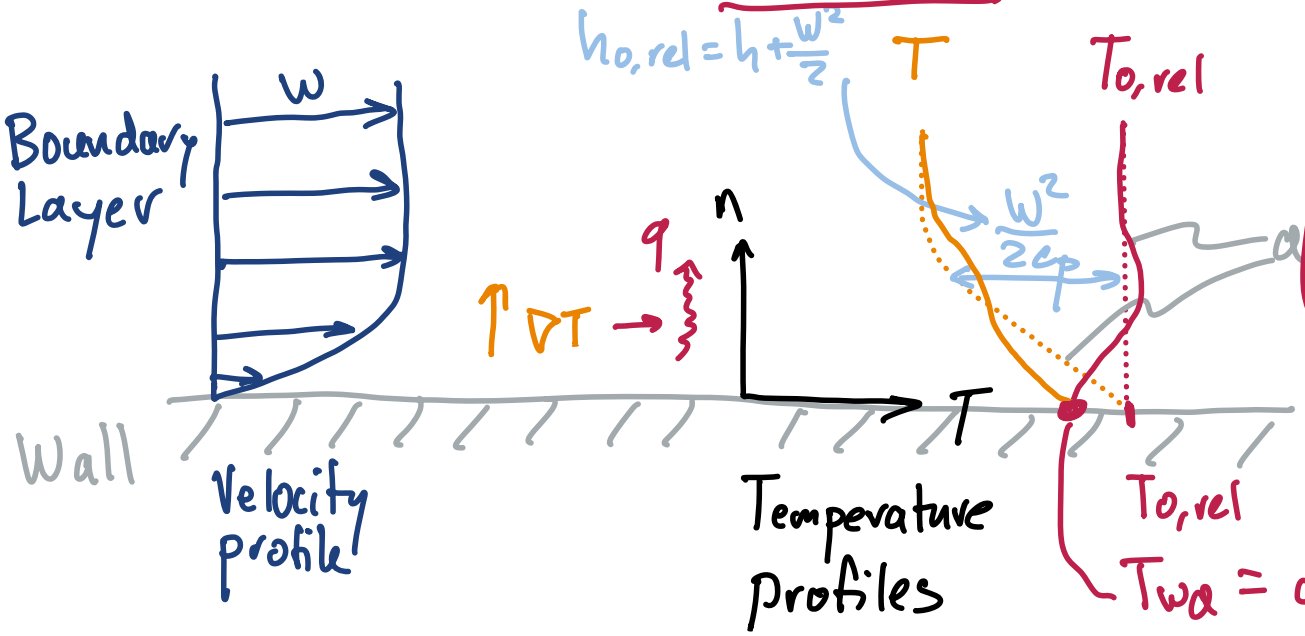


Flow vs. Surface Temperature (Adiabatic Wall)

Assumptions:

- No blade cooling
- No conduction of heat away from blade into disk

Blade surface temperature *approaches* stagnation temperature in frame of blade



$$T_{wa} = T + r_f \frac{w^2}{2c_p}$$

"recovery factor" 0.9–0.95

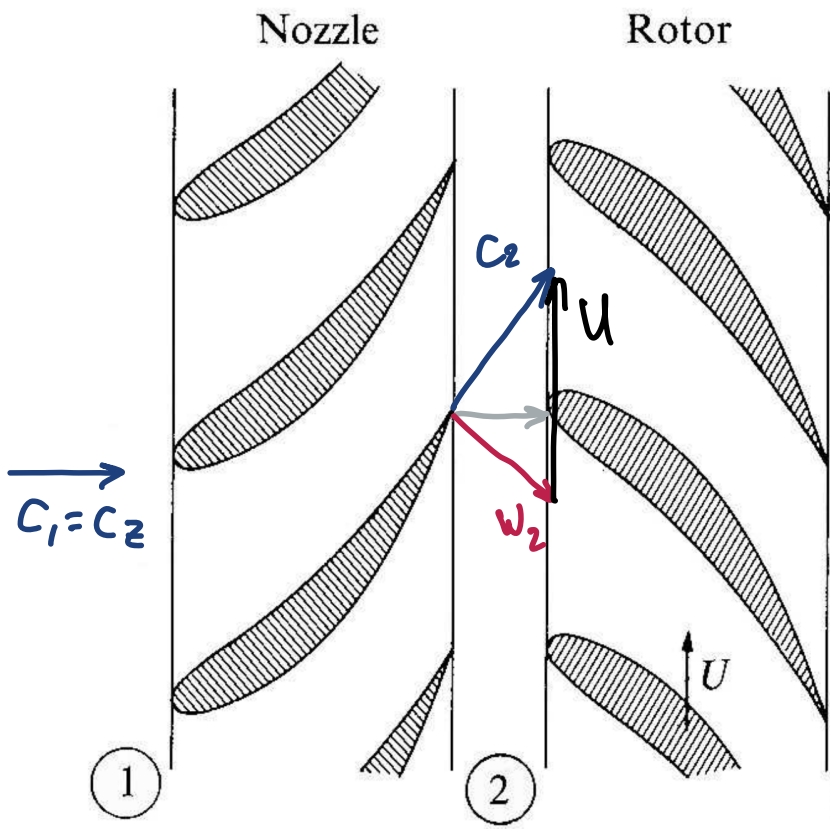
~~adiabatic flow~~

$$T_{wa} = \left(T_{o,rel} - \frac{w^2}{2c_p}\right) + r_f \frac{w^2}{2c_p}$$

$$= T_{o,rel} - (1 - r_f)\frac{w^2}{2c_p}$$

T_{wa} = adiabatic wall temp.

Concept Question: Nozzle vs. Rotor Temperature



Will the equilibrium temperature of the leading edge of an uncooled rotor be greater than, less than, or equal to that of an upstream uncooled nozzle?

Nozzle:

$$T_{wa} = T_1 + r_f \frac{c_1^2}{2c_p}$$

$$= \left(T_{01} - \frac{c_1^2}{2c_p}\right) + r_f \frac{c_1^2}{2c_p}$$

$$= \boxed{T_{01} - (1 - r_f)\frac{c_2^2}{2c_p}}$$

$\gtrless 0?$

Rotor:

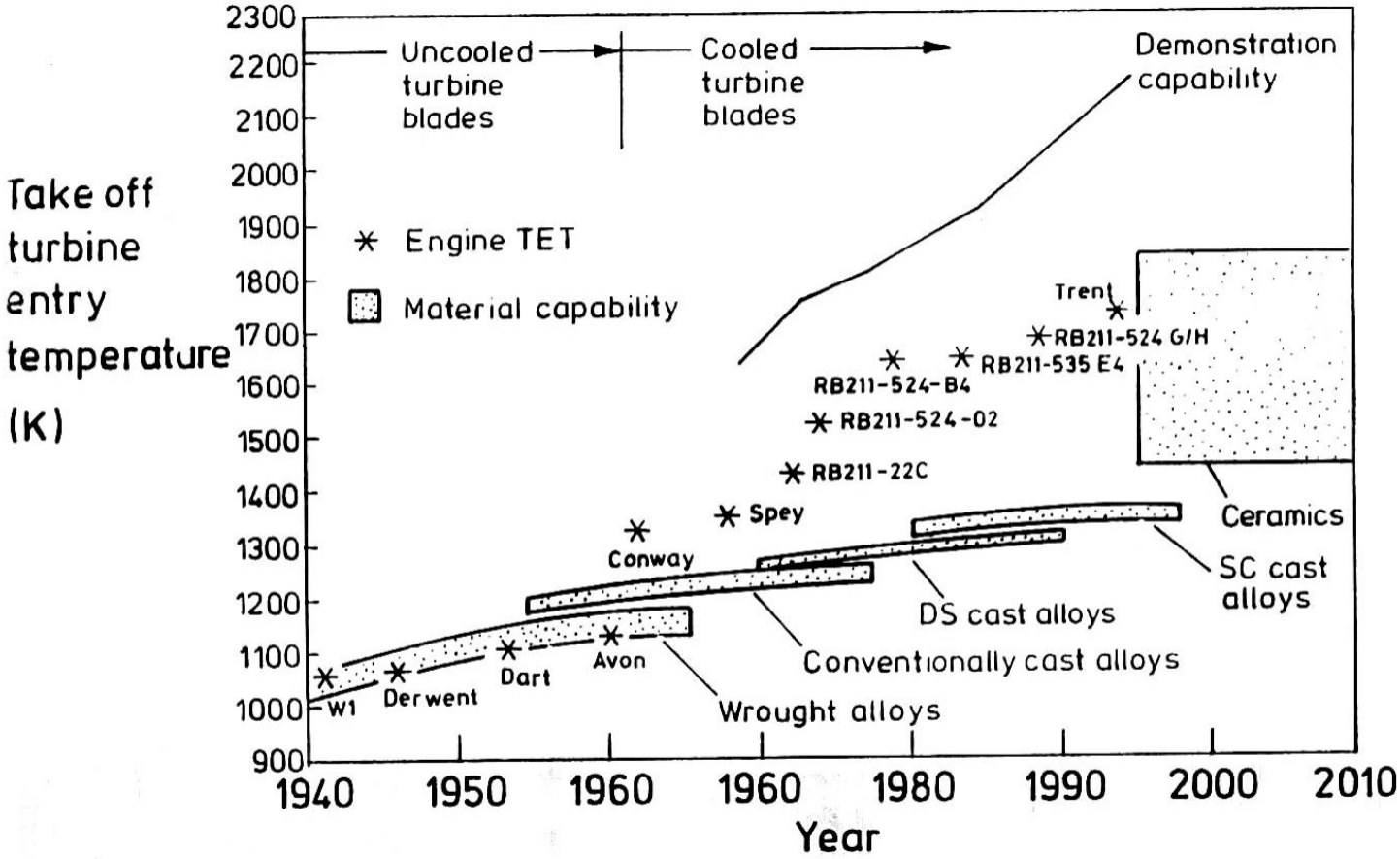
$$T_{wa} = T_2 + r_f \frac{w_2^2}{2c_p}$$

$$= \left(T_{02} - \frac{c_2^2}{2c_p}\right) + r_f \frac{w_2^2}{2c_p}$$

$T_{01} = T_{02}$ (first Law)

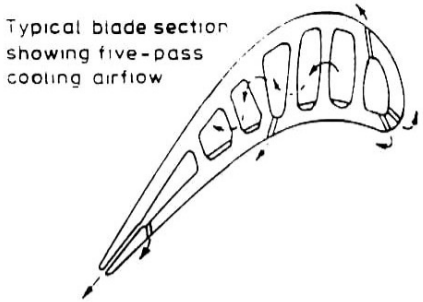
$$= \boxed{T_{01} - (1 - r_f)\frac{c_2^2}{2c_p}} - \frac{c_{02}^2}{2c_p} + r_f \frac{w_{02}^2}{2c_p}$$

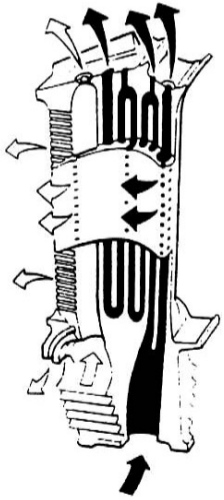
Higher Cycle Temperatures Enabled by Turbine Cooling



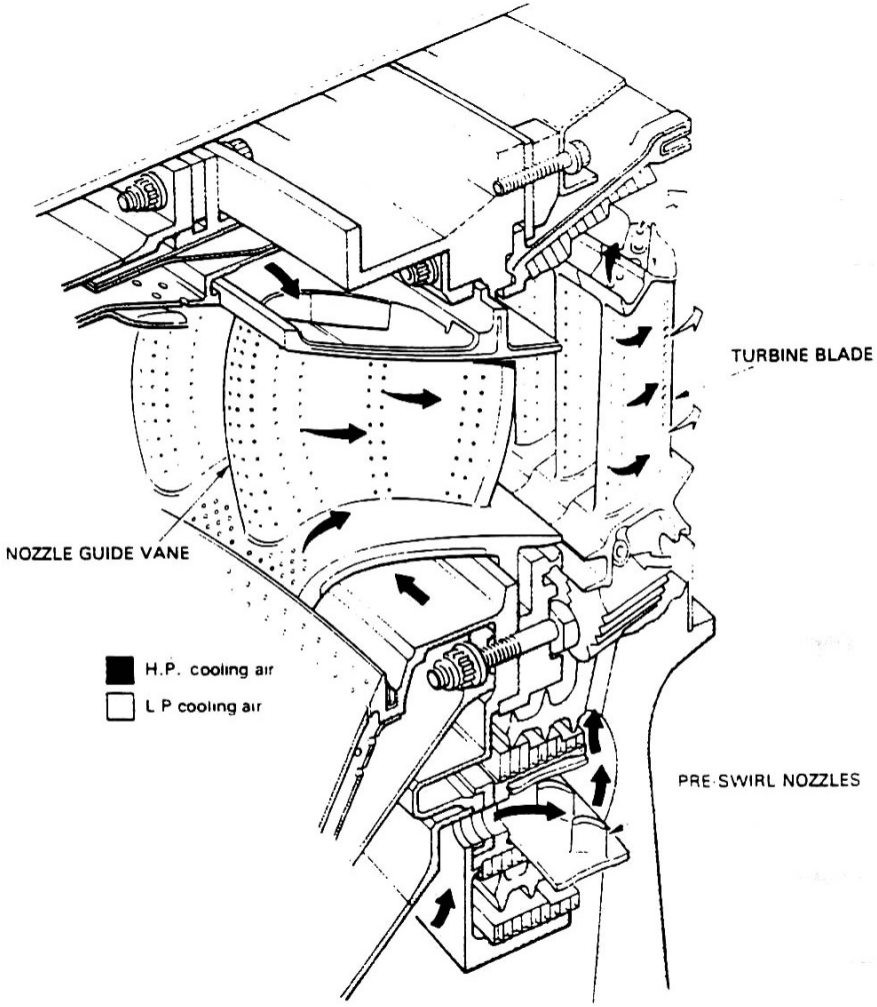
Rolls Royce high-temperature technology advances since 1940

Turbine Convection (Internal), Film (External) Cooling





QUINTUPLE PASS, MULTI-FEED INTERNAL COOLING WITH EXTENSIVE FILM COOLING



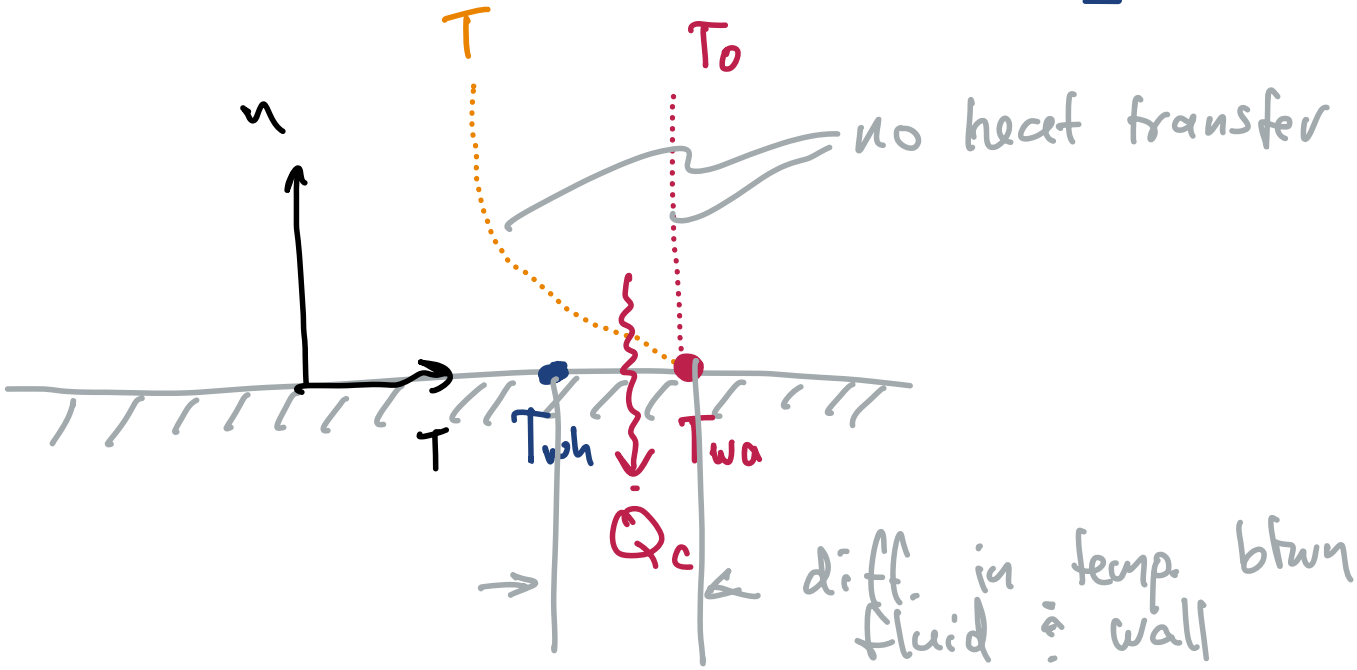
Convective Heat Transfer

Heat flux per unit area (to/from fluid from/to blade) [W/m²]

hot (fluid) temp. @ wall

$$\dot{\mathbb{Q}}_c = h_g(T_{\mathrm{wa}} - T_{\mathrm{wh}}) \tag{11.27}$$

convective heat transfer coeff $\left[\frac{W}{m^2 K}\right]$



Convective Heat Transfer and Stanton Number

- Non-dimensional parameter relating fluid heat transfer and thermal capacity

$$\mathrm{St} = \frac{h_g}{\rho_g c_{pg} w_g} = \frac{\text{convective heat transfer}}{\text{gas flow heat capacity}} \tag{8.27}$$

$$\begin{aligned}
\dot{q}_c &= h_g (T_{wa} - T_{wh}) \\
&= \mathrm{St}\, \rho_g c_{pg} w_g (T_{wa} - T_{wh}) \\
&= \mathrm{St} \frac{\dot{m}_g}{A_g} c_{pg} (T_{wa} - T_{wh})
\end{aligned}$$

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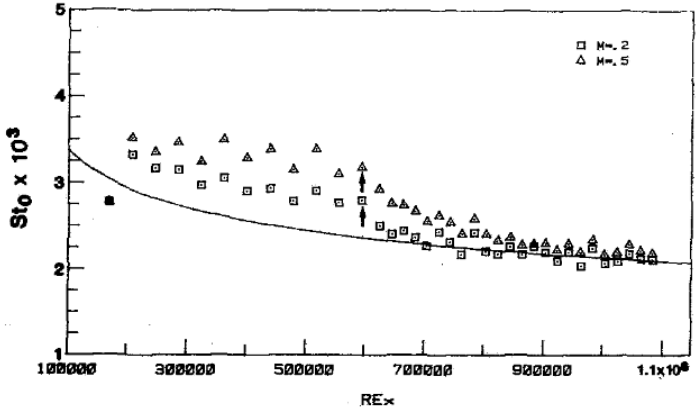
- Empirical correlations for Stanton number

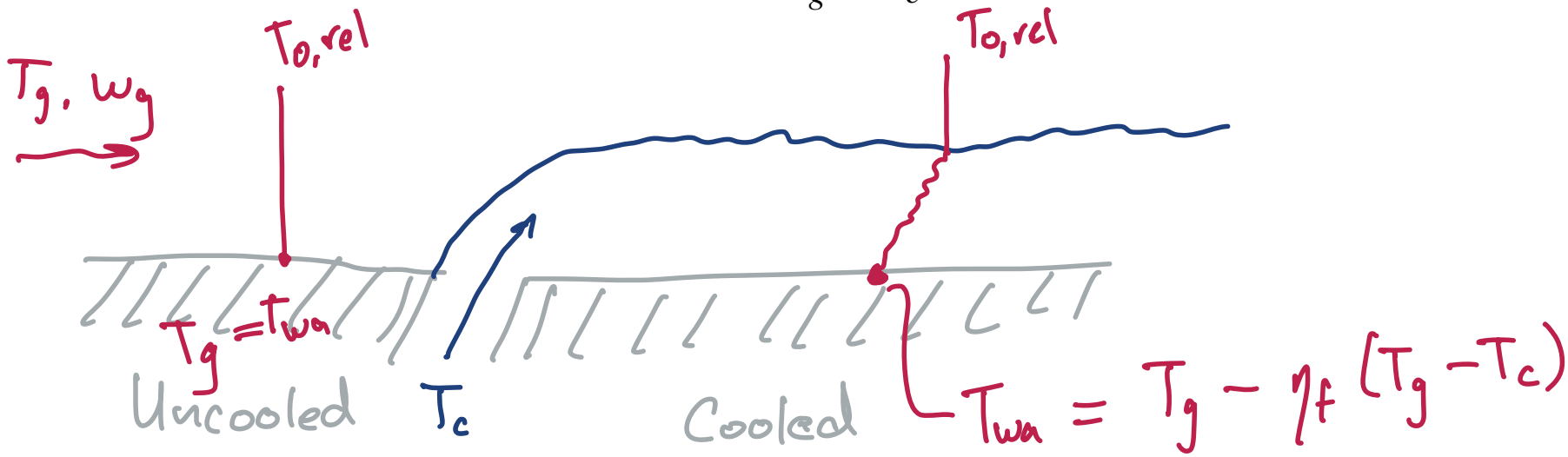
$$\mathrm{St} = f(\mathrm{Re}, \mathrm{Pr}, \mathrm{M})$$



Film Cooling Effectiveness

- Non-dimensional parameter relating hot gas, coolant, and adiabatic wall temperatures

$$\eta_f = \frac{T_g - T_{\mathrm{wa}}}{T_g - T_c} \tag{8.26}$$



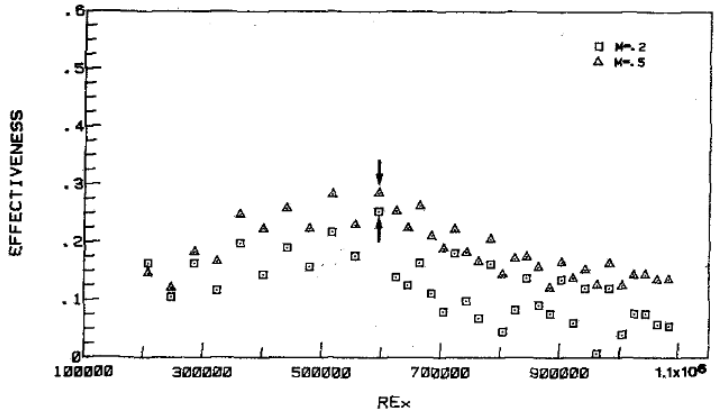
Film Cooling Effectiveness

- Non-dimensional parameter relating hot gas, coolant, and adiabatic wall temperatures

$$\eta_f = \frac{T_g - T_{\mathrm{wa}}}{T_g - T_c} \tag{8.26}$$

- Empirical correlations for film cooling effectiveness

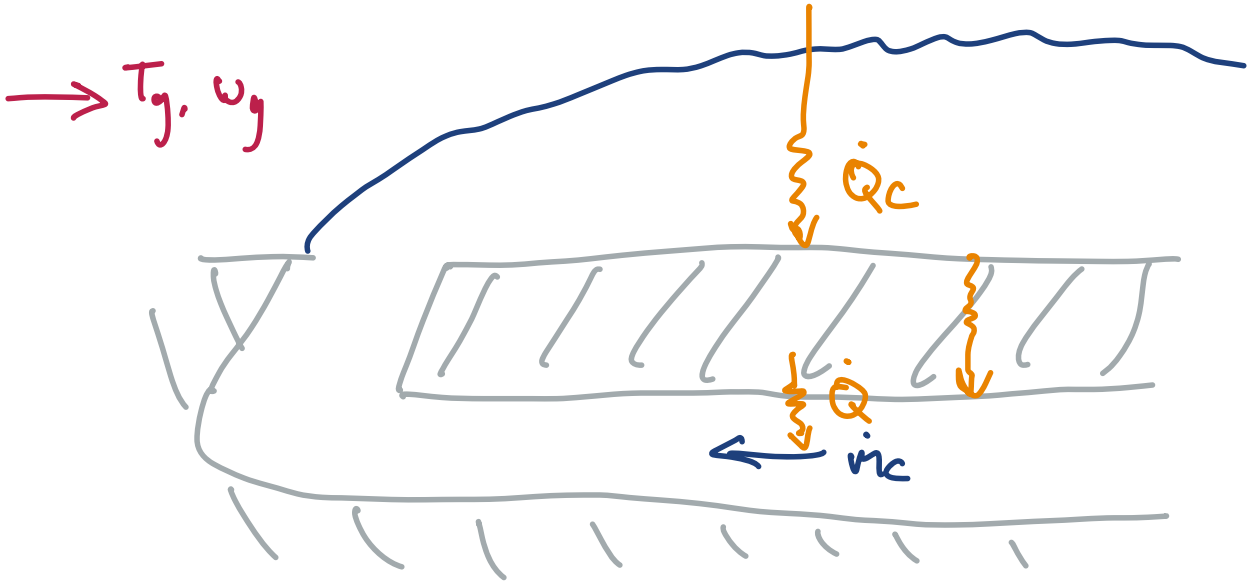
$$\eta_f = f(\mathrm{Re}, \mathrm{Pr}, \mathrm{M}, \text{cooling hole design})$$



Concept Question: Blade Cooling Heat Balance

Consider a turbine blade cooled via internal cooling passages, which lead to cooling holes on the surface for external film cooling

- At thermal equilibrium (i.e., the engine has been running long enough that the flow and blade temperatures are no longer increasing or decreasing with time) how are the heat transfer rates due to external convection and internal convection related?



Example: Blade Cooling Heat Balance

Hot side: $\dot{q}_c = h_g (T_{wa} - T_w)$

blade surface area

$$\dot{Q}_c = St \frac{\dot{m}_g}{A_g} c_{pg} (T_{wa} - T_w) A_s$$

ejected film coolant temp.

Film cooled: $T_{aw} = T_g - \eta_f (T_g - T_c)$

$$\dot{Q}_c = St \frac{\dot{m}_g}{A_g} c_{pg} \left[T_g - T_w - \eta_f (T_g - T_c) \right]$$

initial coolant temp (comp. exit)

Cold side: $\dot{Q} = \dot{m}_c c_{pc} (T_c - T_i)$

Heat Balance: $\dot{m}_c c_{pc} (T_c - T_i) = St \frac{\dot{m}_g}{A_g} c_{pg} \left[T_g - T_w - \eta_f (T_g - T_c) \right]$

Example: Blade Cooling Heat Balance

$$\frac{\dot{m}_c}{\dot{m}_g} = \mathrm{St}\frac{A_s}{A_g}\frac{c_{p,g}}{c_{p,c}}\left[\frac{T_g - T_w}{T_c - T_i} - \eta_f\left(\frac{T_g - T_c}{T_c - T_i}\right)\right] \tag{8.31}$$

Typical values:

Hot gas temperature	$T_g = 1600\ \mathrm{K}$
Blade wall temperature	$T_w = 1200\ \mathrm{K}$
Compressor exit temperature	$T_i = 850\ \mathrm{K}$
Coolant ejection temperature	$T_c = 1100\ \mathrm{K}$
Stanton number	$\mathrm{St} = 0.003$
Film cooling effectiveness	$\eta_f = 0.2$
	$\frac{A_s c_{p,g}}{A_g c_{p,c}} = 10$

$$\Rightarrow \frac{\dot{m}_c}{\dot{m}_g} \simeq 0.036$$

Per blade row

Impact of Turbine Cooling on Performance

- Heat balance used to find required cooling flow rates (see (8.31) derivation)
  - Modern engines may bleed 5-10% of compressor exit air for turbine cooling

- Cycle perspective
  - Effective reduction in T_{04}, relative to combustor exit temperature
  - Cooling enables higher combustor exit temperatures, worth the tradeoff

- Component perspective: account for multiple inlet streams in calculating turbine efficiency

$$\eta_{\mathrm{tt}} = \frac{w_{\mathrm{actual}}}{\sum \alpha_i c_{pi} T_{01i} \left[1 - (p_{02}/p_{01i})^{(\gamma-1)/\gamma}\right]}$$